

Interview Context Features (Sequential Decision Factors)

“These capture the state and history of the interview to guide the next question.”

Answer: We designed these features from well-established assessment and instructional theories. They are not random values; they simulate how an interviewer uses **context** when asking questions. We reviewed literature on *adaptive testing*, *mastery learning*, and *human tutoring*. These sources indicate that interviewers adapt not only to a single answer but also to the sequence of responses. Based on this, we chose context features like **Current Topic**, **Question Difficulty**, **Correct Streak**, and **Wrong Streak**. We then translated these ideas into quantitative rules for our synthetic dataset.

Step 1 – Literature Review

“We first reviewed research on adaptive interviewing, computer-adaptive testing, and educational theory.” The goal was to understand how the interview *trajectory* affects decision-making. In human tutoring and testing, context matters: if a student keeps getting answers right, the tutor will change the approach. If they keep getting them wrong, the tutor will change strategy. We looked at concepts such as:

- **Mastery Learning (Bloom’s Theory):** Once a learner has demonstrated mastery on a topic (by answering several questions correctly in a row), the tutor or system typically moves to the next topic or increases difficulty [5†L2-L7] .
 - **Adaptive Testing:** Standard computer-adaptive tests (like the GRE) adjust question difficulty based on a sequence of correct/incorrect answers [5†L2-L7] . Correct streaks raise difficulty; wrong streaks lower it.
 - **Pedagogical Flow:** Research in intelligent tutoring shows that maintaining an optimal challenge is key. Too many easy questions bored; too many hard ones frustrated. The history of performance informs that balance.
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Step 2 – Extract Contextual Principles

From these studies, we identified key contextual strategies:

- **Maintain or Increase Challenge:** If a candidate is on a roll (long *Correct Streak*) and has been answering *Hard* or *Medium* questions, the interviewer should switch topic or keep difficulty high [5†L2-L7] . This follows *mastery learning* – moving on when the candidate has “mastered” current material.
- **Reduce Challenge:** If the candidate has multiple recent incorrect answers (*Wrong Streak* ≥ 2 or 3), the interviewer should lower difficulty or review basics to build confidence [5†L2-L7] . This avoids frustration (a principle of scaffolding and adaptive testing).

- **Switch Context:** If the candidate has answered several questions correctly on the same topic (*Correct Streak* high), it may be time to shift to a new topic, since remaining on the same topic yields diminishing returns (mastery learning) 【5†L2-L7】 .
- **Question Type & Difficulty Context:** If the current question is already *Hard* and the candidate answers correctly, it may make more sense to switch topic rather than find an even harder question (since our scale might cap out). Likewise, if the candidate is struggling, we might switch to an easier question type (e.g. from application back to theory).

These principles are consistent with educational research on maintaining engagement and flow during learning.

Step 3 – Convert Principles into Rules

Research informs the *strategy*, but not exact numbers. We therefore defined our own thresholds:

- **Mastery Trigger:** “If **Correct Streak** ≥ 3 and current difficulty is *Medium* or *Hard*, then **Switch Topic**.”
Rationale: Consistent success indicates mastery of current topic 【5†L2-L7】 , so move on or change focus.
- **Challenge Increase:** “If **Correct Streak** ≥ 2 , current difficulty is *Easy*, and the candidate keeps answering well, then **Increase Difficulty**.”
Rationale: A short run of correct answers on easy questions means we should up the challenge (adaptive testing principle).
- **Stuck State:** “If **Wrong Streak** ≥ 3 , then **Reduce Difficulty** or **Probe Basics**.”
Rationale: Multiple recent mistakes suggest the current difficulty is too high or there's a gap. We lower the bar (scaffolding principle).
- **Stability:** “If performance is average and no streaks are high, **Maintain Difficulty**.”
Rationale: When there's no clear trend, keep the interview level steady to gather more data.
- **Topic Switch on Hard:** “If a candidate just correctly answered a *Hard* question with a **Correct Streak** ≥ 2 , then **Switch Topic**.”
Rationale: We may have reached the limit on this topic's depth. Bloom's taxonomy suggests broadening scope to application in other topics.

Each rule combines literature-driven strategy with practical thresholds. The educational theory comes from the research, while the exact streak counts and difficulty levels are our design choices.

Step 4 – Generate Dataset

We now create synthetic interview-turn examples using these rules. Each example has both semantic, behavioral, **and** context features, with the next policy label derived from our rules.

Example Row 1:

- Correctness = 94, Coverage = 91, Reasoning = 90, Missing Concepts = 0
- Engagement = 0.91, Confidence = 0.89, Hesitation = 0.08, Eye Contact = 0.92
- **Topic = OOP, Difficulty = Medium, Correct Streak = 3, Wrong Streak = 0**
- **Label (Policy) = Switch Topic**

Interpretation: The candidate is doing very well (semantic scores are high) and has answered 3 in a row correctly on OOP. By **Mastery Learning**, we label the next action as **“Switch Topic”**, even though semantic performance would allow higher difficulty, because we assume the candidate has mastered OOP.

Example Row 2:

- Correctness = 46, Coverage = 42, Reasoning = 39, Missing = 5
- Engagement = 0.42, Confidence = 0.35, Hesitation = 0.76, Eye Contact = 0.41
- **Topic = DBMS, Difficulty = Hard, Correct Streak = 0, Wrong Streak = 2**
- **Label (Policy) = Reduce Difficulty**

Interpretation: The candidate is struggling (low semantic scores) on a *Hard* DBMS question with two recent wrongs. By **Adaptive Testing** and **Scaffolding**, we label the next action as **“Reduce Difficulty”**.

Each dataset row is generated this way: real-looking feature values are chosen so that our rules (derived from educational principles) apply to determine the policy label. This ensures our synthetic dataset is internally consistent and research-driven rather than random.
